

GLPI TICKETING SYSTEM PROJECT REPORT

1. Introduction

The GLPI Ticketing System project was implemented to provide a centralized platform for managing IT support activities within an organization. The system supports the management of users, departments, IT assets, support tickets, service-level agreements, and technical knowledge.

The implemented system separates responsibilities among different user roles. This report focuses on two major operational roles:

- **Administrator (glpi)**
- **Technician (tech)**

The administrator is mainly responsible for configuring and maintaining the environment in which the helpdesk operates, while the technician is responsible for processing support tickets and documenting technical solutions.

The database activity log demonstrates actions including creation of departments and users, configuration of assets and ticket categories, ticket processing, follow-ups, technical tasks, solutions, knowledge-base management, and SLA configuration.

2. Project Objectives

Based on the implemented functions and recorded system activity, the GLPI Ticketing System is designed to support the following objectives:

1. Centralize IT support requests through a ticketing system.
2. Organize users according to organizational departments and entities.
3. Maintain records of IT hardware and software assets.
4. Categorize incidents and service requests for easier management.
5. Allow technicians to monitor and process assigned tickets.
6. Record communication and work performed during ticket resolution.
7. Maintain reusable technical documentation through a Knowledge Base.
8. Apply service-level rules to different ticket priorities.
9. Maintain an activity history for accountability and traceability.

3. Main System Roles

3.1 Administrator

The administrator account in the implementation is `glpi`.

The administrator is responsible for establishing and maintaining the structure of the GLPI environment. Rather than concentrating mainly on individual support incidents, the administrator manages the resources, organizational structure, configuration, and policies required for the ticketing system to operate.

3.2 Technician

The technician account is `tech`.

The technician is responsible for operational helpdesk activities. After users submit support requests, the technician can work with the tickets, record actions performed, communicate through follow-ups, create technical tasks, provide solutions, and document useful information in the Knowledge Base.

This creates a clear separation between **system administration** and **IT support operations**.

4. Administrator Functions and Activities

4.1 Authentication and System Access

The administrator logged into GLPI multiple times during system configuration and management. The logs record successful administrator login events from the system environment.

Authentication ensures that administrative activities are associated with an authorized account.

4.2 Department and Group Management

One of the first administrative activities was establishing the organization's departmental structure.

The administrator created groups including:

- Human Resource Department
- Finance Department
- Sales Department
- Information Technology Department
- Operation Department
- Management Department

The administrator also modified some of these groups after creation.

This structure allows users and tickets to be organized according to their organizational responsibilities.

4.3 User Management

The administrator created multiple employee accounts in the system, including accounts such as:

- `minh.anh`
- `hoang.nam`
- `thu.ha`
- `quoc.huy`
- `ngoc.mai`
- `duc.anh`
- `tuan.khang`
- `van.long`
- `quoc.bao`
- `minh.duc`
- `thanh.dat`

The administrator subsequently updated several user records.

This demonstrates that account creation and maintenance are centralized under the administrative role.

4.4 User-to-Group Assignment

After creating the organizational groups and user accounts, the administrator assigned users to appropriate groups.

This establishes relationships between employees and their departments and allows GLPI to use the organization's structure when handling tickets and permissions.

4.5 Entity Management

The administrator also configured GLPI entities.

The activity log records the creation and configuration of entities such as:

- By Departments (departments other than IT and HR)
- IT
- HR

Users were then associated with appropriate entities.

Entities provide another level of organizational separation and can be used to control resources, users, tickets, and permissions within different areas of the organization.

4.6 Location Management

The administrator configured physical locations, including:

- Reception
- HR Office
- Finance Office
- Sales Office
- IT Room
- Operations Office
- Meeting Room 1

Locations allow tickets and IT equipment to be associated with their physical positions within the organization.

5. IT Asset Management

The GLPI installation was also configured as an IT asset management system.

5.1 Computer Management

The administrator added computers such as:

- HR-LAP-01
- FIN-LAP-01
- FIN-LAP-02
- SALES-LAP-01
- SALES-LAP-02
- SALES-LAP-03
- IT-LAP-01
- OPS-PC-01
- OPS-PC-02
- OPS-PC-03
- OPS-PC-04

The use of standardized asset names helps identify both the department and device.

5.2 Printer Management

The administrator added printers including:

- HQ-MFP-01
 - HQ-MFP-02
 - FIN-PRT-01
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5.3 Network Equipment Management

Network infrastructure was added to GLPI, including:

- HQ-FW-01
- HQ-RTR-01
- HQ-SW-01
- HQ-SW-02

- HQ-AP-01
- HQ-AP-02
- HQ-AP-03

These records represent firewall, router, switch, and wireless access-point infrastructure.

5.4 Monitor and Phone Management

The administrator registered monitors used in Operations, HR, Finance, Sales, and IT departments.

Company phones were also registered, including devices assigned to HR, Finance, Sales, IT, and Operations.

The administrator additionally configured manufacturer and model information, including Samsung and the A55 phone model.

5.5 Software and License Management

The administrator created software inventory records including:

- Windows 11 Pro
- Microsoft 365
- Google Chrome
- Microsoft Teams
- Adobe Acrobat Reader
- 7-Zip
- ERP Client
- CRM
- Payroll
- Accounting Software
- AnyDesk
- PowerShell

Software license records were also added.

This allows GLPI to function not only as a ticketing platform but also as an inventory management system.

6. Ticket Administration

The administrator also performed ticket-level administrative actions.

The logs show that the administrator:

- Created a ticket.
- Deleted individual tickets.
- Deleted multiple tickets using a massive/bulk action.

These actions demonstrate that the administrator maintains higher-level control over ticket records in addition to configuring the environment.

7. SLA and Service-Level Management

The administrator configured service-level settings for different levels of ticket priority.

The system includes service-level entries associated with:

- Critical
- High
- Medium
- Low

The administrator created and later updated SLA/SLM-related records. OLA-related records were also configured.

This establishes a framework in which incidents of different priorities can be associated with different service expectations.

8. Technician Functions and Activities

Unlike the administrator, the technician's recorded activity is focused primarily on support delivery.

8.1 Technician Authentication

The **tech** account logged into the system multiple times to perform helpdesk activities.

The logs also contain some failed login attempts, showing that authentication activity is recorded for both successful and unsuccessful access attempts.

8.2 Ticket Processing

The technician updated multiple support tickets.

Examples in the activity history include updates to tickets 1, 2, 3, 4, 5, and 8.

Updating a ticket allows the technician to maintain the current state of the incident as troubleshooting progresses.

8.3 Ticket Actor Assignment

For ticket 10, the technician performed the action:

Add an actor

Actors represent people involved in the ticket process, such as requesters, technicians, or observers.

This function allows responsibility and participation in a support case to be clearly identified.

8.4 Ticket Follow-Ups

The technician added a follow-up to ticket 10.

Follow-ups provide a record of communication and progress associated with a ticket.

A typical ticket communication process is therefore:

User submits ticket → Technician reviews ticket → Technician adds follow-up → User receives/reads update

This maintains a chronological communication history inside the ticket rather than depending on separate communication channels.

9. Technical Task Management

The technician created several tasks for ticket 9.

The logs contain multiple:

tech adds a task

events.

The technician subsequently updated those tasks several times.

Tasks allow a larger support incident to be divided into individual technical activities. For example, one ticket may require multiple diagnostic, configuration, verification, or testing steps.

The recorded workflow therefore demonstrates:

Create Task → Perform Work → Update Task → Continue Resolution

This provides more detailed tracking than simply changing the overall ticket status.

10. Solution Management

The technician interacted with a solution record associated with a ticket.

The logs show the technician updating a solution object, while user activity later records approval or rejection of proposed solutions.

This suggests the following implemented workflow:

Technician investigates issue → Technician provides/updates solution → User reviews solution → User approves or rejects solution

This is an important part of the ticket lifecycle because it allows the requester to participate in confirming whether the issue has actually been resolved.

11. Knowledge Base Management

Knowledge Base management is one of the most significant technician activities recorded in the system.

The technician created numerous Knowledge Base items, with entries extending through multiple numbered articles.

The technician's recorded Knowledge Base activities include:

- Creating Knowledge Base articles.
- Updating Knowledge Base articles.
- Adding targets to Knowledge Base articles.
- Deleting/purging an article.

The administrator creates and maintains the **Knowledge Base category structure**, while the technician creates and maintains the **technical content** within that structure.

This provides a useful division of responsibility.

For example:

Administrator → Configure Knowledge Base categories

Technician → Create technical articles inside those categories

The Knowledge Base can therefore preserve solutions to previously encountered problems for future reference.

12. Ticket Handling Workflow

Based on the actions recorded in the database, the operational ticket workflow can be represented as:

1. User submits a support ticket
2. Ticket enters the GLPI system
3. Technician reviews and updates the ticket
4. Technician or appropriate actor is associated with the ticket

- 5. Technician communicates through follow-ups**
- 6. Technician creates technical tasks when necessary**
- 7. Technician performs and updates the tasks**
- 8. Technician provides or updates a solution**
- 9. User reviews the proposed solution**
- 10. User approves or rejects the solution**
- 11. Resolved information can be documented in the Knowledge Base**

The administrator supports this process by maintaining the users, organizational structure, categories, assets, and service-level configuration required for the workflow to operate.

13. Functional Architecture of the Project

From the recorded activities, the implemented project can be divided into five main functional areas.

A. User and Organization Management

Managed primarily by the administrator:

- Users
- Groups
- Departments
- Entities
- Locations

B. IT Asset Management

Managed primarily by the administrator:

- Computers
- Monitors
- Printers
- Phones
- Network equipment
- Software

- Software licenses

C. Ticket Management

Operationally handled by the technician:

- Ticket updates
- Actor assignment
- Follow-ups
- Tasks
- Solutions

D. Knowledge Management

Shared responsibility:

- Administrator: Knowledge Base category configuration
- Technician: Knowledge Base article creation and maintenance

E. Service-Level Management

Managed by the administrator:

- SLA configuration
 - SLM configuration
 - OLA configuration
 - Priority-related service levels
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14. Project Evaluation

The recorded activities demonstrate that the GLPI implementation covers more than basic ticket submission.

The system integrates:

- Helpdesk ticket management
- Organizational structure
- User management
- IT asset inventory
- Software inventory
- Technical task tracking
- User-technician communication

- Solution management
- Knowledge management
- Service-level management
- Activity logging

A particularly important design characteristic is the separation of responsibilities between administrator and technician.

The **administrator** builds and maintains the operating environment of the helpdesk.

The **technician** uses that environment to provide IT support and document technical knowledge.

This separation improves organization and prevents operational support tasks from being mixed unnecessarily with system administration responsibilities.

15. Conclusion

The GLPI Ticketing System project demonstrates an integrated approach to IT service management.

The administrator configured the organization, users, assets, ticket classifications, Knowledge Base structure, and service-level settings. These activities established the foundation required for the helpdesk to operate.

The technician then used that environment to perform operational support work. Recorded technician activities include updating tickets, assigning actors, creating follow-ups, managing technical tasks, working with solutions, and creating and maintaining Knowledge Base articles.

Together, these functions create a structured support process:

Administration and Configuration → Ticket Submission → Technician Processing → Communication and Tasks → Solution → User Confirmation → Knowledge Documentation

The project therefore demonstrates how GLPI can be used to centralize IT support operations while maintaining clear role separation, asset information, ticket history, technical documentation, and auditable system activity.